

AMD 6 cores

The upcoming Phenom II X6 CPUs can clock up by 500 MHz when half of the cores are active – this is better than Intel's Turbo Mode

New flagship Lynnfield

Rumours report Intel readying the Core i7 880 (3.06 GHz) for a second quarter launch

DX11

It's been around for a while but do you know all you need to about DX11 and why it's being touted as the next big thing since chocolate icing.

Michael Browne

michael.browne@thinkdigit.com

Though it seems fairly long ago that Microsoft first talked about DX10, for most gamers including some of us at Digit, the scars still remain.

For a farce it seemed – a military-style takeover attempt from DX9 and one that left many users with shiny-new and largely useless DX9 cards; faced with no choice but to upgrade. It also meant upgrading the OS – dependable old Windows XP was ditched in favour of the new, and prodigious resource hog that we know as Vista. No doubt, this made NVIDIA and AMD happy, for this necessitated upgrading to their new DX10 GPUs. So it seems Micro-

soft can bulldoze a product into the market, forcing adoption with its considerable clout in desktop circles. Although this is the way most users view the DX10 fiasco, industry experts have a different opinion. According to them, years of backward compliance was hampering forward growth, which is why the changes made from DX8 and all its revisions to DX9 and all its revisions were subtle enough to be barely noticed. The goal of DX10 was, according to them, to provide a firm foundation for future developments on the DirectX platform, throwing backward compatibility out the window.

There was a lot of trepidation when DX11 was announced – people were wary. This time round, the blow is a slightly softer – Vista also supports DX11. DX11 is, therefore,

not a radical overhaul, but what Microsoft refers to as an incremental upgrade. In fact, so quick was this newcomer brought to market, that one has to suspect the credentials of DX10. From what we've seen, it's clear that DX11 has been designed to combat a lot of the shortcomings of DX10 – that is to say, augment it in every way. And this move is not without a hint of desperation, for the last two years have seen PC gaming giving a lot of ground to consoles. Major development studios and bastions for PC-game development like Epic Games, BioWare and id Software have been working on multi-platform projects from some time. In fact Microsoft claims to have pulled all stops – they even hint that the GPUs in future consoles will, in all likelihood, be based on DX11, much like the graphics hardware in the Xbox 360 and PS3 of today are based on DX9 hardware. DX11 is being touted as *the* DX – not only does it subtly improve on DX10 in nearly all areas, it's also pretty focussed on GPU-accelerated applications – this is something we've been hearing bits and pieces of – but there's nothing really tangible in the market. NVIDIA talks about CUDA while AMD talks about Stream. It's old news that Intel had plans to hop aboard this bandwagon with Larrabee that was supposed to bridge the divide between CPUs and GPUs, but for all market purposes it's gone the way of the dodo.



The difference is clearly visible in S.T.A.L.K.E.R. Call of Pripyat. A DX11 screenshot is on the right, and while they're not identical notice the detail on the cloth near the neck, also the texture on the suit, helmet and gas mask